US selling price baseline results

Recommendation

Keep ASP30 as the reference baseline for the next stage of development. It had the lowest pooled validation MAE and the lowest final holdout MAE and WAPE. The learned alternatives did not deliver a consistent improvement across periods. This is evidence that recent own-product price is a strong predictor for this cohort, rather than evidence that machine learning cannot help.

Final holdout	Result	Meaning
ASP30 MAE	$173.65	Mean absolute price error per product week
ASP30 WAPE	9.71%	Absolute ASP error relative to total actual ASP
ASP30 RMSE	$300.63	Large errors materially affect performance
Scored coverage	566 of 845 weeks	67.0% of candidate product weeks were scored

What the comparison tells us

The weighted blend was effectively tied with ASP30 on holdout: its MAE was only $0.04 higher. Ridge reduced holdout RMSE by 1.29%, but its MAE was $0.61 higher. The nonlinear model had a holdout MAE of $192.14, which was 10.65% worse than ASP30.

The strongest concern is the combination of rising error and lower evaluation coverage in the later period. ASP30 MAE rose from $105.77 in Validation 1 to $173.65 on holdout. Product composition and realized price levels also changed, so this pattern alone does not prove model drift.

Forecast setup and metric definitions

A product week is one product's seven-day forecast and its corresponding realized ASP. Actual ASP is total invoice revenue divided by total invoice quantity within that week. Each scored product week has equal weight in MAE and model fitting; units determine the ASP inside the week.

Model	Forecast logic
ASP30	Use the product's own ASP over the previous 30 days.
Weighted blend	Learn weights for own ASP30, own ASP90 and peer ASP90; weights are nonnegative and sum to one.
Ridge	Add a learned adjustment to ASP30 using the three reference prices, sale recency, invoice row count and active selling days.
Nonlinear	Use small gradient boosted trees to learn an adjustment to ASP30, allowing thresholds and interactions.

Time based evaluation

Stage	Fit through	Requested period	Complete weeks scored
Validation 1	Jan 31 2026	Feb 1 to Mar 31	Feb 1 to Mar 28
Validation 2	Mar 31 2026	Apr 1 to May 31	Apr 1 to May 26

Stage	Fit through	Requested period	Complete weeks scored
Final holdout	May 31 2026	Jun 1 to Aug 31	Jun 1 to Aug 30

Targets use nonoverlapping seven-day blocks, so the final 3, 5 and 1 days of the respective periods are omitted. Only training labels fully observed by the fitting cutoff are used. Parameters stay fixed within a period, while historical inputs update weekly. This tests a refreshed weekly forecast, rather than a single forecast made months ahead.

Model performance across periods

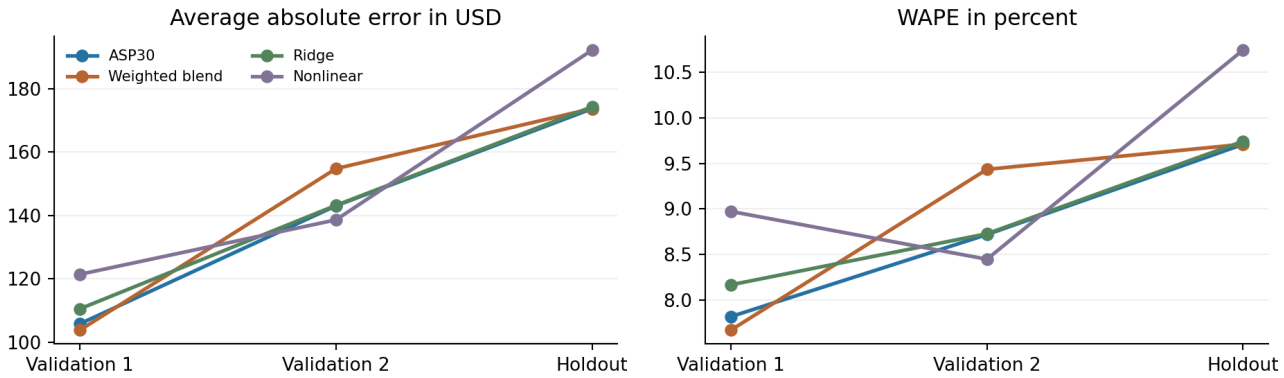


Figure 1 Holdout MAE and WAPE exceed Validation 1 values for all four models. Each model is compared on the same rows within a period.

Period	Model	MAE USD	WAPE	RMSE USD
Val 1	ASP30	105.77	7.82%	160.29
Val 1	Weighted blend	103.78	7.67%	158.04
Val 1	Ridge	110.49	8.17%	161.61
Val 1	Nonlinear	121.37	8.97%	177.16
Val 2	ASP30	143.01	8.72%	246.22
Val 2	Weighted blend	154.73	9.43%	261.79
Val 2	Ridge	143.15	8.73%	246.06
Val 2	Nonlinear	138.57	8.45%	232.14
Final holdout	ASP30	173.65	9.71%	300.63
Final holdout	Weighted blend	173.69	9.71%	301.29
Final holdout	Ridge	174.26	9.74%	296.74
Final holdout	Nonlinear	192.14	10.74%	311.00

Validation 1 favored the weighted blend on all three metrics. Validation 2 favored the nonlinear model. On holdout, ASP30 had the lowest MAE and WAPE, while Ridge had the lowest RMSE. A single favorable fold therefore does not establish a durable advantage.

Using pooled validation MAE, ASP30 scored $124.84 versus $127.21 for Ridge, $129.87 for the blend and $130.18 for the nonlinear model. ASP30 was selected before the holdout was evaluated. WAPE and MAE rank models identically within a period because they use the same rows and actual-price denominator.

What the learned models reveal

The blend increasingly relies on recent own price

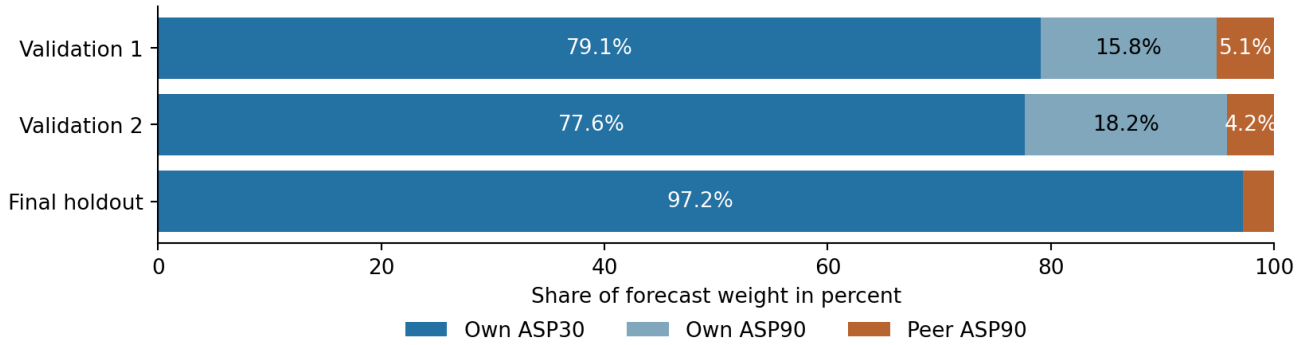


Figure 2 Training fitted weights shift strongly toward ASP30 in the final refit. Final peer weight is 2.8%; ASP90 is effectively zero.

Fitting stage	Own ASP30	Own ASP90	Peer ASP90
Validation 1	79.07%	15.78%	5.15%
Validation 2	77.63%	18.17%	4.20%
Final holdout	97.20%	0.00%	2.80%

In the final fit, the blend is approximately 97.20% ASP30 plus 2.80% peer ASP90. That makes its near tie with ASP30 unsurprising. Peers receive a small weight in every fit. Broad family and type groups may include products at very different price levels, so peer ASP can pull a forecast away from the product's own price. This is a possible explanation to test, not a demonstrated cause.

Ridge provides a small tradeoff rather than a clear win

The selected Ridge alpha was 1000, the strongest regularization among the five tested values. Larger alpha shrinks adjustment coefficients more strongly after feature scaling. Its selection suggests that restrained adjustments generalized better within this tested grid; it does not establish that 1000 is an optimal universal setting.

On holdout, Ridge RMSE fell from $300.63 to $296.74, while MAE rose from $173.65 to $174.26. This is consistent with reducing some large misses while slightly worsening ordinary errors. The 1.29% RMSE gain is small and has no uncertainty interval, so it does not justify changing the MAE-based selection rule after seeing the holdout.

The nonlinear model is unstable across periods

The nonlinear model improved Validation 2 MAE by 3.10% relative to ASP30, but worsened Validation 1 MAE by 14.75% and holdout MAE by 10.65%. Its added flexibility did not generalize reliably. Overfitting, changing sales patterns and omitted explanatory factors are hypotheses for further diagnosis; the metrics alone cannot distinguish them.

Coverage and limits of the evaluation

Accuracy applies to product weeks with a realized target and every required historical feature. All four models use the same comparison rows, so their scores are directly comparable within each period. The coverage percentage is an evaluation eligibility rate, not a production forecast availability rate: future sales determine whether an ASP target exists.

Period	Candidate weeks	Scored weeks	Coverage	Products scored
Validation 1	520	406	78.1%	63
Validation 2	520	426	81.9%	63
Final holdout	845	566	67.0%	61

Period	No future sales	Missing ASP30	Missing peer ASP90
Validation 1	106	25	0
Validation 2	87	19	0
Final holdout	272	82	0

These exclusion counts overlap. In the holdout, 272 weeks had no future sales and 82 lacked ASP30; they must not be added together. The net exclusion was 279 of 845 candidate weeks. No candidate week lacked peer ASP90, so peer availability was not the reported coverage bottleneck.

Sparse products need a separate interpretation

The holdout scored 61 products, compared with 63 in each validation fold. The candidate counts correspond to a 65-product cohort across 8, 8 and 13 weekly blocks. Six of the 61 scored holdout products had fewer than four evaluated weeks. Their individual scores are useful for investigation but too thin to support confident product-specific model selection.

Data definitions matter

Own history and target prices use the selected transactions, while the intended peer reference uses other products in the full US Enterprise source. If the selected file omits transactions for a product, the forecast describes that transaction subset. Confirm transaction completeness and retain the full-source file version used for the run.

The earlier missing pricing_date field also needs a documented resolution. Confirm that the date assigned to the full source matches the selected source and reflects when an invoice was observable. The exported metrics cannot independently establish that date mapping or verify the original full-source transactions.

Zero-sales weeks have no realized selling price and should be reported separately. Products without recent own sales also need an explicit operational fallback; they cannot be treated as covered simply because a peer reference exists.

Where the baseline makes its largest errors

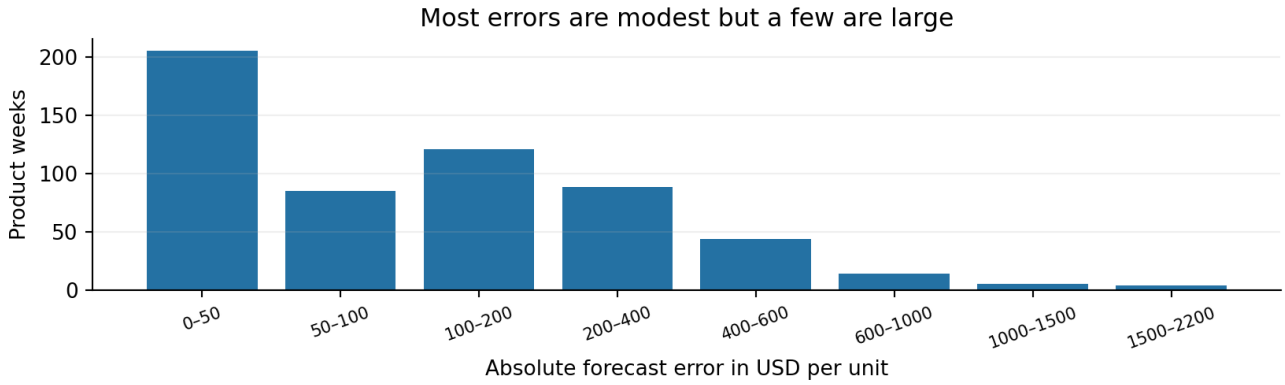


Figure 3 ASP30 holdout absolute errors have a long upper tail. Median error is $93.38, below the $173.65 mean.

For ASP30, 90% of scored holdout errors were at or below $439.41 and 95% were at or below $561.97. The maximum error was $2,177.83. The ten largest errors account for 14.97% of total absolute error, which helps explain why RMSE is much higher than MAE.

Product ID	Scored weeks	MAE USD	Share of total error
98DG935PJZ	13	731.28	9.67%
10TBMCG2LC	11	470.09	5.26%
36H9DOF8CS	11	437.53	4.90%
16VT6348MT	12	330.92	4.04%
99R3052SJU	11	285.79	3.20%

These five products contribute 27.07% of ASP30 total absolute holdout error. Start the transaction-level review here: examine price jumps, customer or channel mix, quantities, contract changes and invoice timing. Those factors are investigation targets rather than explanations established by these exports.

The baseline tends to underpredict in the holdout

Average prediction minus actual ASP was −$54.15, indicating an average underprediction. Mean actual ASP also rose from $1,352.60 in Validation 1 to $1,788.71 on holdout. These are cohort-level observations; shifts in the evaluated products and weeks can contribute to both patterns.

Winning more products does not guarantee lower pooled error

The weighted blend improved MAE on 38 of 61 products, Ridge on 26 and the nonlinear model on 17. Yet ASP30 won pooled holdout MAE. The size of each gain or loss and the number of scored weeks per product determine the pooled result. Equal-product average MAE slightly favors the blend ($175.84) over ASP30 ($176.31), but that was not the prespecified selection metric and should remain a sensitivity check.

Recommended next steps and decision criteria

Keep ASP30 as the comparison standard

Use the product's previous 30-day ASP as the forecasting baseline in the next iteration. Retain the blend and Ridge as challengers. The current evidence does not support replacing the baseline with the nonlinear model or claiming a meaningful difference between ASP30 and the blend.

Resolve data and coverage questions first

Document the invoice date definition, source versions, selection rules and whether selected product histories are complete. Add a coverage view that separates no-sales target weeks from weeks where a forecast could not be produced because own history was missing. Define a fallback policy for products with insufficient recent sales, and evaluate that policy separately.

Investigate the largest errors before adding complexity

Review the five high-contribution products and the largest individual misses. Compare errors by historical transaction count, active days, recency, product family and price level. Use these analyses to decide which additional predictors are justified and available at forecast time. For example, known contract or promotion changes may be useful if their availability can be established.

Test improvements on a future untouched period

If changing peer groups, lookback windows, model features or regularization, tune those changes using development data. Reserve a new later period for a fresh evaluation because June–August has now been inspected. Keep pooled MAE as the main criterion unless the business explicitly chooses another objective in advance.

For a proposed replacement, require a repeatable MAE improvement, acceptable RMSE and coverage, and no material deterioration in important product segments. Estimate uncertainty using paired comparisons that preserve common calendar weeks across products. Do not infer a reliable win from a few cents of average error difference.

What these results can support

The results support a baseline forecasting decision for this selected US Enterprise cohort under weekly refreshed historical inputs. They do not establish optimal prices, causal price elasticity, profit gains or the quality of the original product-selection strategy. A cohort chosen using later sales information also limits generalization to newly selected products.